

Collective Behavior of Place and Non-place Neurons in the Hippocampal Network

Highlights

- A successful unified theoretical framework for population states
- Maximum entropy model predictions have high precision agreement with data
- Network interactions explain a substantial amount of population activity in CA1
- Place cells and non-place cells encode information collectively

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In Brief

Correlation patterns in CA1 hippocampus only partially arise from place encoding. Meshulam et al. utilize a population-level modeling approach to uncover collective patterns of activity in CA1 neurons that substantially reflect not only position but also their internal network state.



Collective Behavior of Place and Non-place Neurons in the Hippocampal Network

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SUMMARY

Discussions of the hippocampus often focus on place cells, but many neurons are not place cells in any given environment. Here we describe the collective activity in such mixed populations, treating place and non-place cells on the same footing. We start with optical imaging experiments on CA1 in mice as they run along a virtual linear track and use maximum entropy methods to approximate the distribution of patterns of activity in the population, matching the correlations between pairs of cells but otherwise assuming as little structure as possible. We find that these simple models accurately predict the activity of each neuron from the state of all the other neurons in the network, regardless of how well that neuron codes for position. Our results suggest that understanding the neural activity may require not only knowledge of the external variables modulating it but also of the internal network state.

INTRODUCTION

The hippocampal formation plays a key role in various memory processes and is crucial for spatial navigation. In humans, individual hippocampal neurons whose activity is associated with specific items or episodes were found sparsely among a larger population of non-responsive neurons (Gelbard-Sagiv et al., 2008; Quiroga et al., 2005; Waydo et al., 2006). In rodents, cells coding for place, velocity, and head direction have been recorded from the hippocampus and its neighboring structures (Hafting et al., 2005; McNaughton et al., 1983; Sargolini et al., 2006). One well identified group of hippocampal neurons are place cells (O'Keefe and Dostrovsky, 1971). These cells are active selectively whenever the animal visits a particular location; i.e., each of these neurons has a specific place field in the environment. Typically, when the animal is familiar with a given environment, about 30% of CA1 pyramidal neurons will fire with significant place fields, and the activity of these cells constitutes

a cognitive map (O'Keefe and Nadel, 1979) that spans the full environment. The remaining cells, often referred to as silent cells (Thompson and Best, 1989), fire with little to no spatial selectivity and usually are less active overall (Kubie and Muller, 1991; Leutgeb et al., 2004; O'Keefe and Conway, 1978; O'Keefe and Nadel, 1979; Thompson and Best, 1989). Although some neurons qualify inarguably as place cells with strong, reliable place fields, for others, the boundary between "place" and "non-place" in a given environment is arbitrary.

One way of interpreting the phenomenology of place cells is that individual cells respond independently to the location of the animal at any moment in time. This approach stems from the view of pyramidal cells responding to specific combinations of complex features of the sensory environment. However, activity in CA1 seems to be more complex than a simple evoked response to external stimuli and cannot all be accounted for by this "sensory features" view. A different approach is to treat the cells as an interacting network, asking how their collective behavior can give rise to place-modulated activity. The latter idea has been widely employed in theoretical models of the hippocampus, which suggest attractor dynamics as the potential mechanism underlying three of the main functions of hippocampus: memory storage, pattern separation, and pattern completion (Fuhs and Touretzky, 2006; Káli and Dayan, 2000; Redish and Touretzky, 1999; Samsonovich and McNaughton, 1997).

A large fraction of hippocampal studies focus on sub-regions CA1 and CA3, and in particular on the sequential activity exhibited by their neurons' spatial (and non-spatial) fields. While it has been suggested that CA1 contributes to the order of a spatial memory sequence, it is not yet known whether CA1 is involved in the integration of sequential elements of a memory or in bridging events that happen sequentially during an episode (Farovik et al., 2009; Hunsaker et al., 2008; Kesner et al., 2005). Although many of the models for hippocampal circuitry refer to CA3 as the main attractor network in the hippocampus (Battaglia and Treves, 1998; Colgin et al., 2010; Jensen and Lisman, 1996; Levy, 1996; Lisman, 1999; Treves and Rolls, 1992), other computational studies suggest that, even though CA1 has less recurrent connections than CA3, it might still function as a continuous attractor, maintaining the continuity of sequential activity (Droulez and Berthoz, 1991; Tsodyks and Sejnowski, 1995).

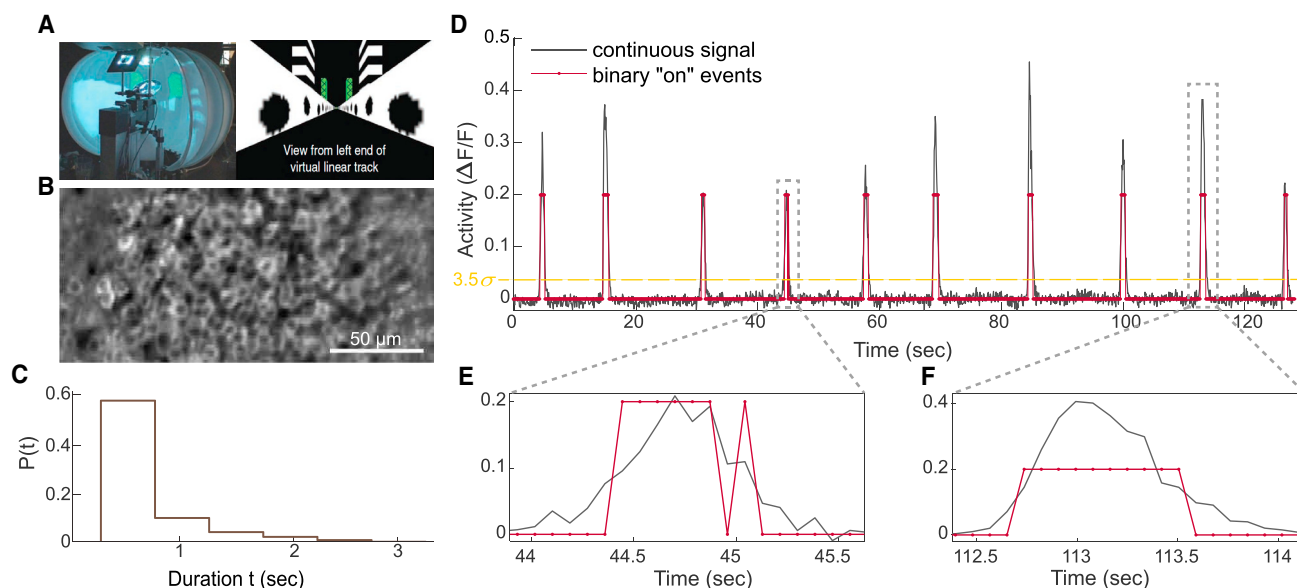


Figure 1. Experimental Setup

(A) Left: a photograph of the experimental apparatus, consisting of a spherical treadmill, a virtual reality apparatus, and a custom two-photon microscope. Right: a top view of the virtual environment of a linear track. (Adapted from Dombeck et al., 2010).

(B) Field of view under the microscope. Scale bar, 50 μm .

(C) Distribution of "on" activity events durations. The majority of events are ~ 500 ms long.

(D) Identifying significant calcium transients. Shown in gray is the raw signal from a neuron in units of normalized fluorescence. Shown in red (arbitrary height) is the binarization of the signal. The neuron is on (which is converted to a "1" value) at every time point where the red line is not 0. The neuron is off (which is converted to a "0" value) at every time point where the red line is 0 (see Method Details in the STAR Methods). The dashed yellow horizontal line shows the 3.5σ threshold in the distribution of all fluorescence values. Points above this threshold qualify in our initial step of binarization of the signal as potential activity events.

(E and F) Enlarged examples of two raw activity transients (gray) and their binarized versions (red).

Although some experimental studies have tried to characterize the instability in the place code and to probe the activity of cells whose activity is not significantly place-modulated (Epsztein et al., 2011; Fenton and Muller, 1998; Ferguson et al., 2011; Kelemen and Fenton, 2013), little to no theoretical effort has been dedicated to studying the phenomenology of CA1 beyond position encoding. Furthermore, recent development of large-scale neural imaging technologies now allows us access to the activity of all neurons, place- and non-place-coding, in a field of view. With this progress arises the need for theoretical approaches that will help elucidate these large population-level dynamics.

Among the possible mathematical models that can reproduce a given pattern of neural activity, we chose the one that does so without incorporating additional structure or assumptions. This minimally informative model is the one that maximizes the entropy of the system (Jaynes, 1957, 1982). In this study, we use the maximum entropy approach to build a model for the probability of all different joint activity patterns of the neurons. We report a successful description of the full probability distribution inferred solely from mean activities and pairwise correlations, which yields accurate predictions both for higher-order phenomena in the network and for the activity of individual place and non-place neurons in relation to the rest of the network. Finally, we show that this collective description, for the population as a whole, yields better predictions than the classical view of individual place cells as independently encoding locations on a spatial map.

RESULTS

The Data: Cellular Resolution Imaging of Neural Activity in the Mouse during Runs along a Virtual Track

We analyze data taken from transgenic mice that express a genetically encoded calcium indicator, GCaMP3. Our virtual reality setup allows imaging of neural activity with cellular resolution in awake head-restrained mice while they run on a spherical treadmill (Figure 1A). Each imaging session included up to ~ 80 simultaneously active neurons in the CA1 hippocampus, recorded as a mouse runs along a 4-m-long virtual linear track. Right before the end of the track, the mouse received a water reward. Runs were consecutive in nature because when the mouse reached the end of the virtual reality environment, the next one started immediately. In agreement with the literature, sub-populations of the imaged neurons (usually $\sim 30\%$) were found to be place cells, with activity in specific place fields along the environment. (Figure 2C; Figure S4).

The raw data from this experiment are essentially a movie. To reduce this to activity of individual neurons as a function of time, we follow the pre-processing steps described in Method Details in the STAR Methods. In an outline, we correct the movies for motion of the brain, identify each neuron as a "region of interest" in the movie, verify these regions manually, and then associate the activity of each cell with the integral of the fluorescence

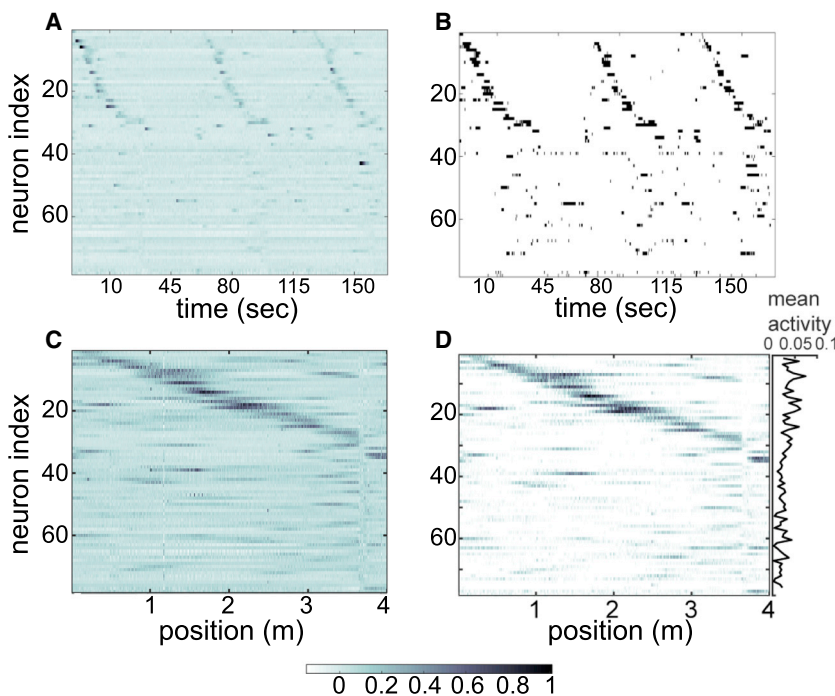


Figure 2. Place and Non-Place-Related Activity

(A) Three consecutive runs down the linear track. During each traversal of the environment, 32 of the 78 cells imaged in the field of view exhibit place-modulated activity. Place cells were sorted based on the averaged activity's center of mass (C). This is the thresholded continuous fluorescence signal as in Figure 1C.

(B) Binarized version of (A). The sequential nature of the place activity is preserved during the discretization process.

(C) Continuous neural activity averaged across runs, calibrated against position, and sorted based on center of mass.

(D) Binarized neural activity averaged across runs, calibrated against position. Right: mean activity values corresponding to the neurons on the left. Non-place cells are usually less active than place cells but rarely silent.

The Model: Maximum Entropy

If we limit ourselves to describing each cell as being active or silent, $\sigma_i = 1$ or 0, respectively, for each neuron i , then a network of N neurons has $\Omega = 2^N$ possible states. With $N = 78$ cells in our experiment, this number of possible states is $\Omega \sim 3 \times 10^{23}$, a genuinely astronomical number. We expect that

signal over the corresponding region. We normalize the fluorescence traces extracted from each neuron and set a threshold above which all maxima identified are recognized as potential activity events; this threshold is set to $\sim 3.5\sigma$, where σ is the SD of the distribution of all fluorescence values of that session (dashed horizontal yellow line in Figure 1D). Next, we discard transients whose shape contradicts a rising-decaying pattern typical of neural activity (see Method Details in the STAR Methods). Finally, we binarize the time series: a neuron state σ_i is defined as 1 for every time bin during an identified activity event and as 0 for the rest (and most) of the time, when it is silent, as shown in Figure 1D. Because each point in our field of view is scanned every 70 ms, we use the same discretization as our effective "time bin" to binarize the signal. Figures 1E and 1F depict two events and their binarized versions. A closer look at Figure 1E reveals a case where, although the event was clearly identified as significant activity, our algorithm found a part of the decay slope that best fits a burst of activity followed by a single event rather than one unified burst of activity because the fluorescence level falls too quickly to be explained by continuous firing. Although the temporal resolution obtained through imaging is too low to resolve single spikes, the loss of information is less significant than it might be in other systems because CA1 neurons tend to burst, not to fire single spikes, and GCaMP3 captures these bursts reliably.

Analysis was performed on all concatenated frames from single sessions, which included any time period when the mouse was running as well as the inter-trial intervals and any time points where the mouse was not engaged in the task, if such existed (rare). Data were used as a single stream, and no behavior-related information was incorporated into the model other than for validation or visualization purposes.

neural activity is organized, and so the network does not wander at random among all of these possibilities. Although there are many ways in which we could try to characterize this organization, perhaps the simplest approach is to ask for the probability distribution over all the possible states, $P(\{\sigma_i\})$: how likely is it that we will find any particular combination of active and inactive neurons at any one moment in time?

A probability distribution over Ω states is just a list of Ω numbers that sum to one. In large networks, it is not possible to conceive an experiment that would allow us to measure all of these numbers. Thus, any attempt to make a "probabilistic model" of neural activity must rest on some (dramatically) simplifying hypotheses. As emphasized long ago by Jaynes (1957), the maximum entropy construction provides a path for building models in which simplifying hypotheses are explicit and testable.

The idea of the maximum entropy method is to build models that match certain experimental facts exactly but otherwise have as little structure as possible. Thus, if we draw states out of the probability distribution, then these combinations of activity and silence across this network will be as random as they can while still matching, on average, the experimental facts we have chosen as being crucial. The choice of what to match embodies our intuition, and the shorter the list of facts, the more drastic our simplification. We emphasize that there is no obviously correct choice for which facts to match: each choice represents a different simplifying hypothesis, and must be tested, quantitatively.

In trying to choose which facts to match, we are looking for a relatively small set of things we can measure reliably that are sufficient to capture the way in which the patterns of activity in the network are ordered. As an example, we can get a very accurate model for the joint distribution of velocities of all the birds in a flock by matching the correlations between individual birds

and their near neighbors; the global ordering of the flock, the correlations among fluctuations in the velocities of distant birds, and even higher-order correlations among multiple birds all emerge from propagation of the near-neighbor correlations, as predicted by the maximum entropy model (Bialek et al., 2012, 2014).

Previous work on maximum entropy approaches to the description of neural activity has explored different possibilities for what should be matched. At one extreme, the network might be characterized by the fact that certain specific combinations of activity and silence occur with surprisingly high frequency, and we could match these frequencies for the most common combinations (Ganmor et al., 2011b). At the opposite extreme, we could focus on measures of global activity and match the frequency with which we see K out of N neurons being active simultaneously without regard to their identity (Tkačik et al., 2013). Here we pursue the original idea of matching the mean activity of individual neurons and the correlations between pairs of neurons (Schneidman et al., 2006; Tkačik et al., 2006, 2009).

We expect that networks with different mean levels of activity will behave differently, and so we insist that any model of the network as a whole match the average activity of each neuron. For each neuron i , we have $\sigma_i = 1$ when that neuron is active, and $\sigma_i = 0$ when it is silent; matching the mean activity of each cell is the statement that, if we compute the mean in our model probability distribution $P(\{\sigma_i\})$, then we get the same answer as from the experiment:

$$\sum_{\{\sigma_i\}} P(\{\sigma_i\}) \sigma_i = \langle \sigma_i \rangle_{\text{expt}}, \quad (\text{Equation 1})$$

where $\langle \dots \rangle_{\text{expt}}$ is the average computed from the experimental data. Similarly, we know that correlations between pairs of neurons capture some aspects of the network's coherent activity, and so we will test the hypothesis that matching these correlations is sufficient to predict all of the possible higher-order structures. Mathematically, the pairwise correlation between neuron i and neuron j can be defined either as a covariance,

$$C_{ij} = \langle (\sigma_i - \langle \sigma_i \rangle) (\sigma_j - \langle \sigma_j \rangle) \rangle, \quad (\text{Equation 2})$$

or as a correlation coefficient,

$$c_{ij} = \frac{C_{ij}}{\sqrt{C_{ii} C_{jj}}}. \quad (\text{Equation 3})$$

To match this entire matrix of correlations, we need to be sure that our model matches the matrix of second moments,

$$\sum_{\{\sigma_i\}} P(\{\sigma_i\}) \sigma_i \sigma_k = \langle \sigma_i \sigma_k \rangle_{\text{expt}}. \quad (\text{Equation 4})$$

There are infinitely many probability distributions that are consistent with the constraints in Equations 1 and 4; among these, we want to choose the model that has the least possible structure or equivalently generates the most random possible states. Although such a characterization may seem vague, Shannon (1948) proved that the only consistent way to measure the (lack of) structure in a probability distribution, or the degree of randomness, is to compute the entropy of the distribution,

$$S = - \sum_{\{\sigma_i\}} P(\{\sigma_i\}) \log[P(\{\sigma_i\})]. \quad (\text{Equation 5})$$

Concretely, then, we want to find the probability distribution that maximizes S while obeying the constraints from Equations 1 and 4. The formal solution to this constrained optimization problem is

$$P(\{\sigma_i\}) = \frac{1}{Z} \exp[-E(\{\sigma_i\})]. \quad (\text{Equation 6})$$

$$E(\{\sigma_i\}) = - \sum_{i=1}^N h_i \sigma_i - \frac{1}{2} \sum_{i,j=1}^N J_{ij} \sigma_i \sigma_j. \quad (\text{Equation 7})$$

We note that this is mathematically equivalent to the Boltzmann distribution for a set of Ising spins σ_i that experience external magnetic fields h_i and interact with one another through couplings J_{ij} (Mézarard et al., 1987). The partition function Z serves to normalize the distribution,

$$Z = \sum_{\{\sigma_i\}} \exp[-E(\{\sigma_i\})]. \quad (\text{Equation 8})$$

This formal solution for the probability distribution involves parameters, $\{h_i, J_{ij}\}$, and there are exactly as many of these parameters as we have constraints from matching the mean activity and pairwise correlations that we see in the population; in principle, then, the parameters of our model are determined exactly by solving Equations 1 and 4. In practice, there is no simple analytic form for these exact expectation values, and so we compute these by Monte Carlo simulation. Thus we “solve” Equations 1 and 4 by using Monte Carlo to generate an estimate of the left side of the equation, comparing with the right side (from the data), and then adjusting the parameters until we get closer to equality; for details, see Figure S1. Our strategy, going from raw fluorescence data to the model, is summarized in Figure 3.

Maximum Entropy Model for a Population of Hippocampal Neurons

We apply the maximum entropy approach to the experimental data on a population of 78 CA1 neurons. The crucial experimental quantities are the mean activities of each neuron, shown in Figure 4A, and the pairwise correlations, shown in Figure 4E. Because we have described activity by a binary variable, the mean activity measures the fraction of time bins in which a neuron is “on,” and we see that this is quite small (~ 0.03), as expected. Corresponding to these low probabilities, the parameters h_i in our model (Equation 7) are relatively large and negative.

As noted in Figure 2, a significant fraction of cells in the population are place cells. For these, we expect positive correlations between cells with overlapping place fields and negative correlations between cells with non-overlapping place fields. More generally, we see that the distribution of correlation coefficients peaks and has the bulk of its weight, at c_{ij} , small and negative (Figure 4G), and the covariance matrix of the activity is dominated by small negative terms far from the diagonal (Figure 4C). Corresponding to these experimental results, the coupling constants J_{ij} in the model also have a distribution peaking at small negative values

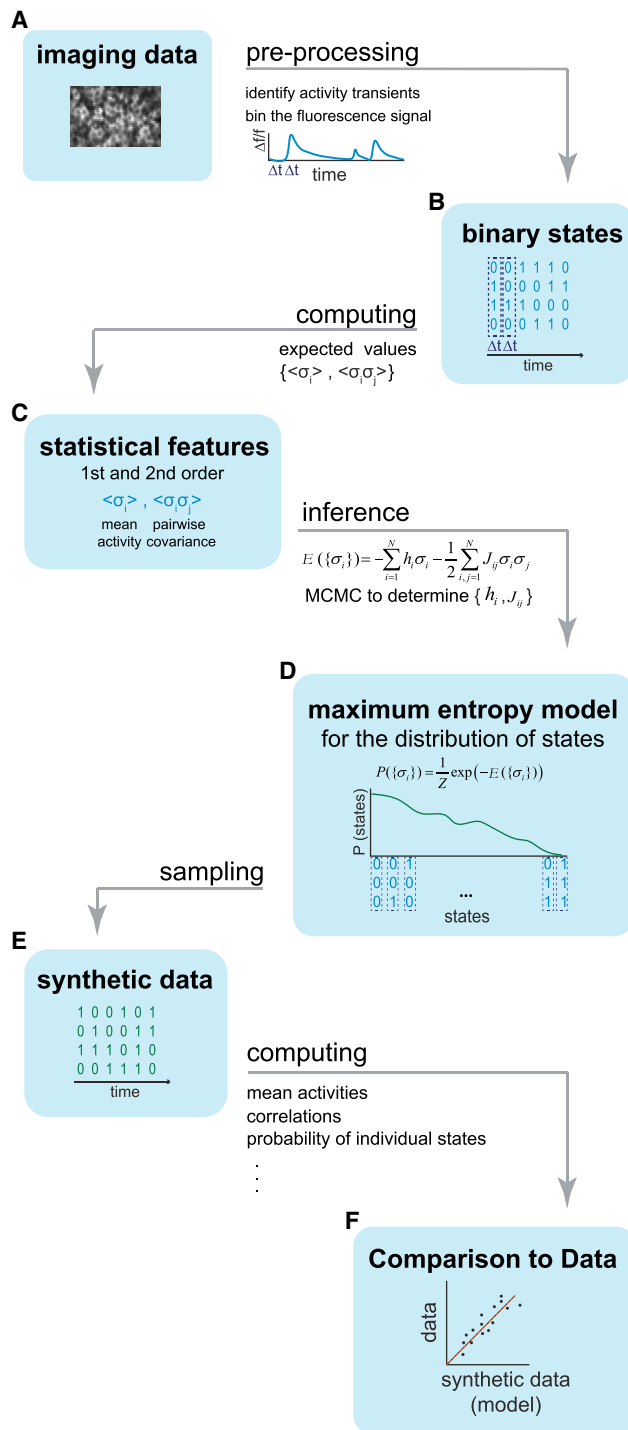


Figure 3. Steps in Building the Maximum Entropy Model

(A) A typical field of view in the CA1 hippocampus. Initial steps include motion correction and identification of individual cells.

(B) After discretizing the signal from each neuron, we now have a matrix of all concatenated trials for all neurons where each neuron was assigned a “0” or “1” value for every moment in time.

(C) Computing the statistical features of the data to which we are going to fit the model. We require the model to match exactly the first and second moments.

(Figure 4F), and when we view these coupling as a matrix, we see widespread, weakly negative terms away from the diagonal, with a small set of positive terms clustered near the diagonal (Figure 4D).

The basic properties we see in these data are reproduced in five other datasets from different mice running through the same virtual track (Figure S2). There are uncertainties in the parameters but no sign that we are overfitting because learning the model from a limited fraction of the data captures mean activities and correlations that agree with those computed from the remainder of the data (Figure S1).

Model Predictions

Maximum entropy models are, by definition, the least structured models that can match particular experimental facts. But the set of facts we choose to match—here, the mean activity and pairwise correlations of the neurons—is small compared with the space of possible states for the network, and there is no guarantee that these few measured quantities provide a sufficient characterization of the system. What is crucial, however, is that everything else we can predict about the behavior of the network from the model involves no new parameters. Thus, what follow are predictions, not fits.

Although the mean probability of a single cell being active is small (Figure 5A), there are many neurons in the population, and so it is possible to observe not just a single cell being active but many cells being active simultaneously. Although we do not have enough data to estimate the probability of every distinct “word” built out of combinations of activity and silence, we can estimate the probability $P_N(K)$ that any K neurons out of the N cells in the network (any K , regardless of their identity) are active in the same small window of time. As shown in Figure 5A, there is significant probability out to $K \sim 8$, and we can follow the small probabilities out to $K = 12$. Across most of this range, the model makes correct predictions within the (small) error bars of the measurement.

The model we have defined predicts the probability of every possible state the network could occupy. It is natural to think not about the probabilities themselves but about their (negative) logarithm, which measures how surprised we should be by each particular pattern of activity (Shannon, 1948); in the equivalent Boltzmann distribution, this is the “energy” assigned to each state (Equation 7). The model makes a prediction for the distribution of energies, but we can also walk through the data, compute the energy of every state we see, and thus estimate the distribution of energies in the data. Although states with more cells active are less likely (Figure 5A), there are more ways of choosing these active cells out of the population, leading to a non-trivial distribution of energies, shown in Figure 5B. We see that theory and

(D) Finding the values of a set of h_i and J_{ij} that match the statistical features of the data via Markov Chain Monte Carlo simulation of the model. The result is a full probability distribution so that every possible population state is assigned a specific probability to occur.

(E) Sample population states from the inferred probability distribution to obtain a matrix of synthetic data.

(F) To test whether the inferred distribution is a good model, we compare the same measures computed on the real data to those computed on the synthetic data.

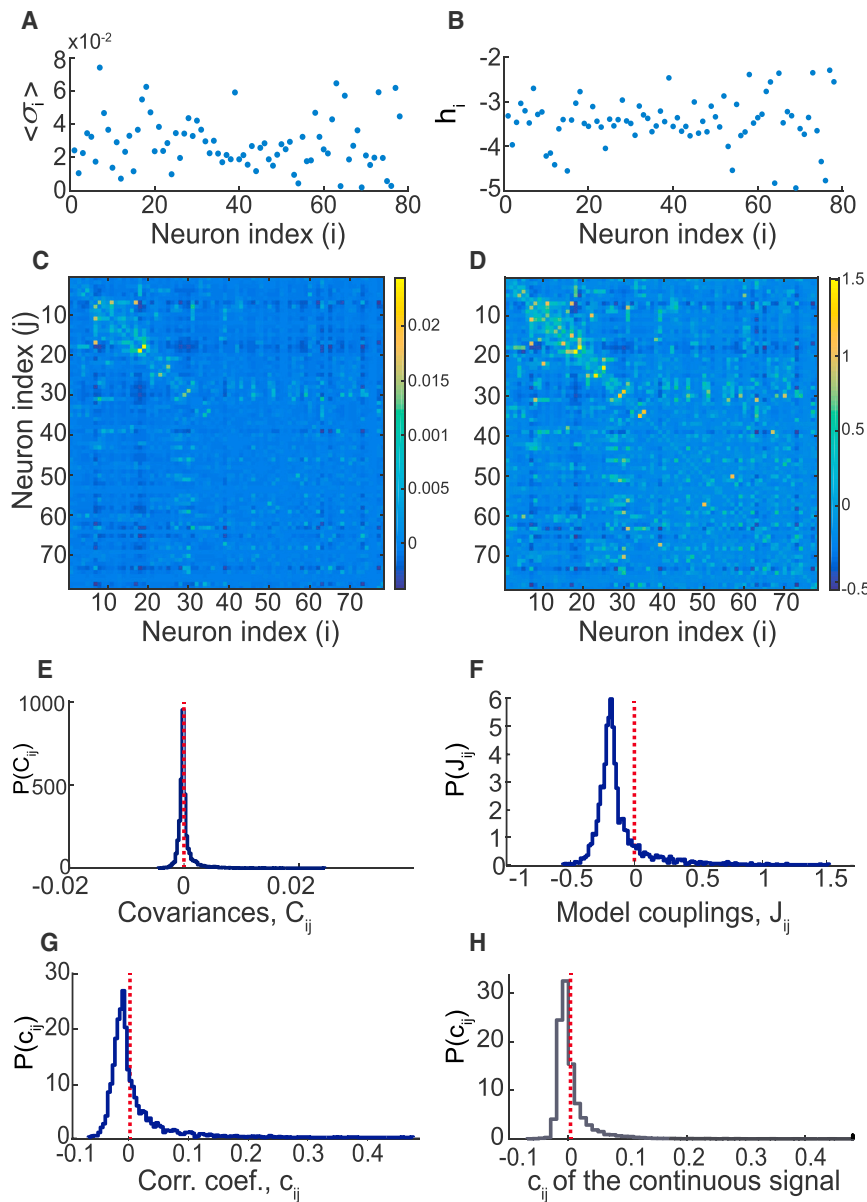


Figure 4. Learning the Maximum Entropy Model

(A) Mean activity, $\langle \sigma_i \rangle_{\text{expt}}$, for each one of the neurons, computed from the data.
 (B) Coefficients, h_i , (magnetic fields) in the model; positive values bias the cell to be active.
 (C) Pairwise covariances, C_{ij} , as defined in Equation 2, computed from the data. C_{ii} was set to 0 for ease of visualization.
 (D) Coupling constants of the model, J_{ij} . Positive couplings favor positively correlated activity. J_{ii} is redundant with h_i and is set to zero.
 (E) Probability distribution of the covariances in the data, C_{ij} for $i \neq j$, as defined in Equation 2. Note the peak at slightly negative values, indicating a significant population of neurons being weakly negatively correlated, as expected from place cells, whose firing is mostly orthogonal to each other.
 (F) Probability distribution of the coefficients (coupling constants), J_{ij} , obtained after fitting the model.
 (G) Probability distribution of the correlation coefficients in the data, c_{ij} for $i \neq j$, as defined in Equation 3.
 (H) Probability distribution of the correlation coefficients, c_{ij} , as defined in Equation 3, in the continuous version of the data, after event detection (baseline set to 0 and detrended) but before binarization. The binarization process preserves the general structure of the correlations in the data.

These correlations are small but significantly non-zero for most of the distinct 7.6×10^4 triplets in the population. Figure 6A shows that the model correctly predicts these correlations, within error bars, across the full dynamic range of the data. To examine the quality of our predictions more closely, we zoom in on the bulk of small ($|C_{ijk}| < 4 \times 10^{-4}$) correlations and compare the error in our predictions with the experimental error in measuring these triplet correlations. Concretely, we make small

experiment agree very well out to energies $E \sim 25$. This energy is equivalent to a probability of $e^{-E} \sim 10^{-11}$, corresponding to events that should occur once every few hundred years. Although, of course, we cannot know whether we have correctly predicted this tiny probability for any single state, what we are seeing is that the model correctly predicts the number of these rare states, each of which is seen only once in the data.

Because we build our model out of the pairwise correlations among neurons, it is natural to test the predictions for higher-order correlations. The probability that K out of N neurons are active, as in Figure 5A, is one combination of all of the higher-order correlations, but we would like to look in more detail. To do this, we check the correlations among triplets of neurons,

$$C_{ijk} \equiv \langle (\sigma_i - \langle \sigma_i \rangle) (\sigma_j - \langle \sigma_j \rangle) (\sigma_k - \langle \sigma_k \rangle) \rangle. \quad (\text{Equation 9})$$

bins along the C_{ijk} axis, and within each bin, we compute the root-mean-square difference between predicted and measured correlations as well as the root-mean-square error in our measurements, which we can think of as error bars on predictions and measurements, respectively. We see in Figure 6B that the two measures of error are essentially equal across a wide range of correlation values: our model predicts the triplet correlations with a precision that essentially matches the experimental error; one cannot do better.

For neurons in the hippocampus, one obvious source of correlations is the existence of place fields: cells with overlapping place fields should have positive correlations in their activity, and cells with distant place fields must be anticorrelated. We certainly see something of this pattern in the data of Figure 4C, and it is tempting to think that all correlations could be

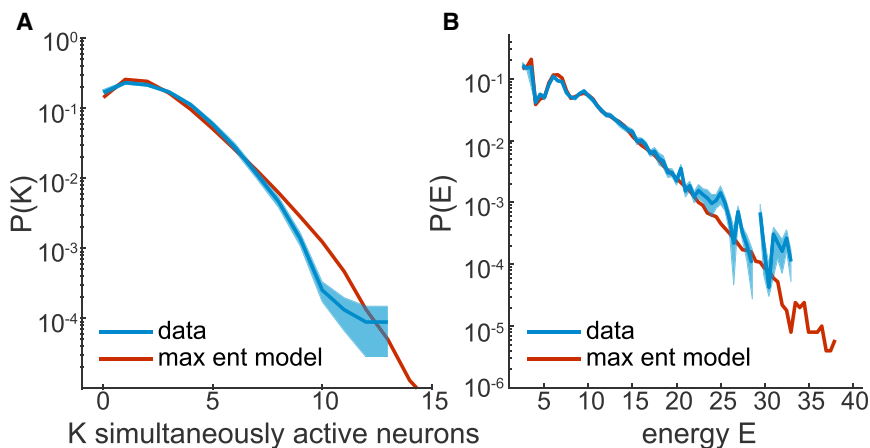


Figure 5. Model Predictions

(A) The probability that K of the $N = 78$ neurons in the population are active simultaneously. To obtain these probabilities, we pool together all states where any K neurons are active together while the rest of the neurons are silent, comparing the data (blue) and the model (red). Error bars show SD across random halves of the data.

(B) The distribution of effective energies, or log probabilities, the model assigns to every possible state of the network. Shown in blue is the distribution over states as they occur in the experiment. Shown in red is the distribution predicted from the model itself. Error bars show SD across random halves of the data.

understood in this way, in which case our success in predicting triplet correlation may be less surprising. To explore this possibility, we constructed a model in which cells respond independently to the position of the animal along the virtual track. Given the probability of each cell being active as a function of position, as in Figure 2D, the model of “independent place cells” has no free parameters (see Method Details in the STAR Methods), and, in particular, we can predict correlations among pairs or triplets of neurons. The prediction of individual pairwise correlations based on place fields alone is not bad (Figure S3A), but the prediction of global properties of these correlations, such as their eigenvalue spectrum (Figure S6D), is worse, and the prediction of triplet correlations is much worse (Figure S3B).

In Figure 6C, we group triplets of neurons into bins based on their observed correlation and plot the mean and SD of predictions in each bin from both the maximum entropy model and the independent place field model. For the maximum entropy model, this is just a different way of looking at the results in Figure 6A, and, correspondingly, we see that the predicted correlations form a tight band around the observed correlations. But for the independent place cell model, predicted correlations span a much smaller dynamic range, and for the bulk of triplets with small $|C_{ijk}|$, there is almost no systematic relationship between predictions and observations. These results suggest strongly that correlations in the network are not simply a response to the environment but are most accurately described as a collective property of the network itself.

We further investigated whether this collective behavior can be accounted for by a few dominant components that might exist in the network beyond place encoding. Among the natural candidates are a global network excitation signal and instability in place coding. As reported in Examining the Collective Behavior of the Network in the STAR Methods and shown in Figure S6, these cannot account for the correlation structure the network exhibits beyond place encoding. It seems, then, that this population level property is not a result of any simple, low dimensional signal we expect to see in the hippocampal network and that it stems from both place and non-place cells.

If activity in a network is collective, then we should be able to predict the activity of each individual neuron from the pattern of activity in the rest of the network. As noted above,

the model we are exploring is mathematically equivalent to a model for spins in a magnet, and thus the influence of the network on a single neuron can be summarized by an “effective field”:

$$h_i^{\text{eff}} = E(\sigma_1, \dots, \sigma_i = 0, \dots, \sigma_N) - E(\sigma_1, \dots, \sigma_i = 1, \dots, \sigma_N) \quad (\text{Equation 10})$$

$$= h_i + \sum_{j \neq i} J_{ij} \sigma_j \quad (\text{Equation 11})$$

Positive effective fields favor neurons being active; more precisely, the probability of the single neuron i being active is given by the logistic function

$$P(\sigma_i = 1 | h_i^{\text{eff}}) = \frac{1}{1 + \exp(-h_i^{\text{eff}})}. \quad (\text{Equation 12})$$

Thus, the model predicts the probability of any single neuron being active at any moment in time, given the state of all the other neurons in the network.

To test the predictions of the model, we walk through the data and, at each moment, for each neuron, we compute the effective field given the state of all the other neurons (Equation 11); we also mark whether the target neuron was active. In Figures 7A and 7B; we focus on two particular cells (indexed 14 and 77 in Figure 2) and plot, as a function of time, the probability that a single neuron should be active, given the state of all the other neurons in the population; we emphasize again that the model makes an unambiguous prediction and that there are no free parameters to adjust. The predicted probability consists of a series of relatively brief peaks separated by periods of probability near zero, and this is reproduced in the data, where we see that these neurons are active only during the predicted peaks. The neuron shown in Figure 7A has a clear place field in this environment (Figure 7C), and the peaks of activity correspond to times when the mouse passes (virtually) through the place field. Importantly, the model we have built makes no reference to the position of the mouse. Thus, to the extent that we are able to predict the activity of place cells, this is because information about place is represented collectively in the network. In contrast, the activity of the neuron

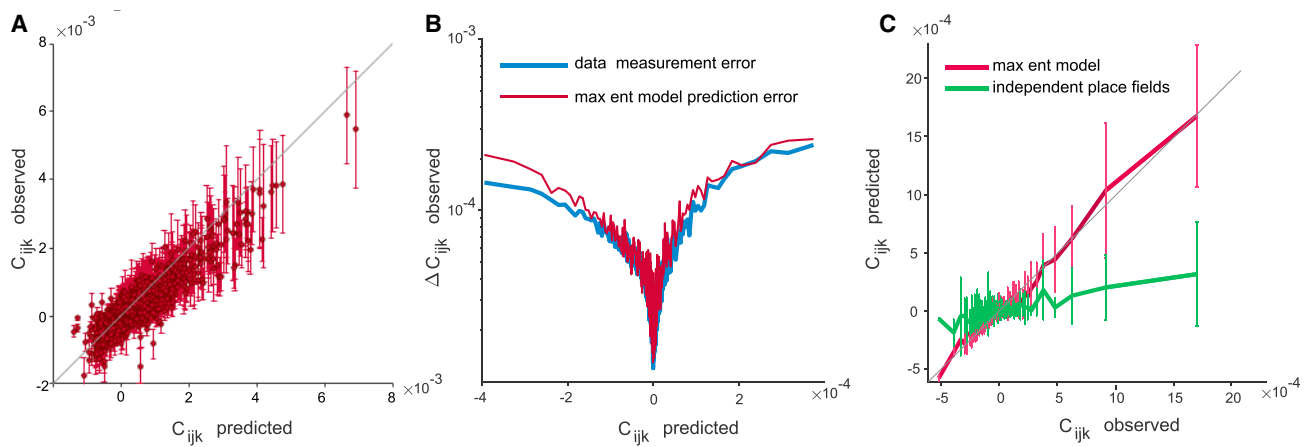


Figure 6. Triplet Correlations

(A) Predicted versus observed triplet correlations, C_{ijk} , as defined in Equation 9.

(B) Comparison of the maximum entropy model prediction errors for individual triplet correlations and the measurements errors computed from the data itself. The two sets of errors are on the same scale and are especially similar for the more common, smaller correlation values.

(C) Predictions of triplet correlations from our maximum entropy model (red) and from an independent place field model (green). The x axis is binned so that values represent a small range of 3-cell correlations rather than individual ones. The animal's location can only poorly account for the higher-order structure in the data. The maximum entropy model captures the higher-order structure much better.

shown in Figure 7B is hardly spatially modulated in this environment, yet the model still predicts brief periods of high probability for the cell being active, separated by longer periods of near zero probability, and again the actual moments of activity follow these predictions quite well. Thus, it appears that both place cells and the remaining neurons that do not code for a particular location are part of the same patterns of collective activity.

The effective field h_i^{eff} (Equation 11), which determines the probability of a single neuron being active, can be thought of as having three components. First, each neuron has an intrinsic bias h_i , independent of all the other cells in the network. Activity in other neurons adds to this bias through the terms $\propto J_{ij}$ in Equation 11. These latter terms can be further separated into contributions from place cells and from non-place cells. In Figure 7E, we sum up the magnitude of total contributions, $|J_{ij}|$, for each cell from the two groups: from place cells in purple and from non-place cells in yellow. The place cells are the first, shaded, 32 columns; the remaining are non-place. In Figure 7F, we show a scatterplot of these last two contributions for each activity event in which the neuron is on, separating again the case where the target cell i is itself a place cell or a non-place cell. Perhaps the most important conclusion is that contributions from the two classes of cells are comparable, both overall and for each individual event across the epochs: our ability to predict the activity of one cell from the patterns of activity in the rest of the network depends both on neurons with a clear place field in the environment as well as on those that are not strongly place-modulated, more or less equally. This reinforces the conclusion that, regardless of the level of place modulation in their activity, all cells are part of the same patterns of collective activity.

If the effective field correctly predicts the probability of a neuron being active, then, conversely, a neuron being active should predict a high value of the effective field. Thus, if we

compare moments when a cell is active or inactive, we should see very different distributions of the predicted probability, and this is shown in Figure 7G. Pooling all of these data together across time, we can plot the actual probability of a cell being active as a function of its effective field and compare this with the prediction of the model (Equation 12). The results of this comparison, shown in Figure 7H, show that theory and experiment agree very well across the full dynamic range. We emphasize that these results hold true independent of the specific definition of which neuron is a place cell and which is not. Our model predictions remain valid for any level of location encoding of the cell, whether significantly, partially, or hardly at all.

In addition to treating all neurons in the population on an equal footing, we also looked into moments where place cells fail to exhibit their typical place-modulated activity. In Figure 8A, we show the probability predicted from our model for each of the place cells in the population during two successive runs along the virtual track. Figure 8B shows, for comparison, the neural activity for the same two runs. The first sequence is an example for a run with “missed” fields (neurons 21–25), whereas, in the second sequence, all place cells are active at the expected times. We see that our model of collective activity in the network manages to capture accurately even those moments where a place cell misses its place field. For the second run, where all neurons were active, the model predicts no missed fields. In contrast to the success of our model for collective activity, treating the cells as having independent place fields necessarily predicts the same pattern of activity during every run along the track (Figure 8C). Any variations must be ascribed to “noise” in the system. In contrast, we find that a cell missing its place field is not an individual deviation disconnected from the rest of the population but, rather, that the network state is predictive of the place field being dropped.

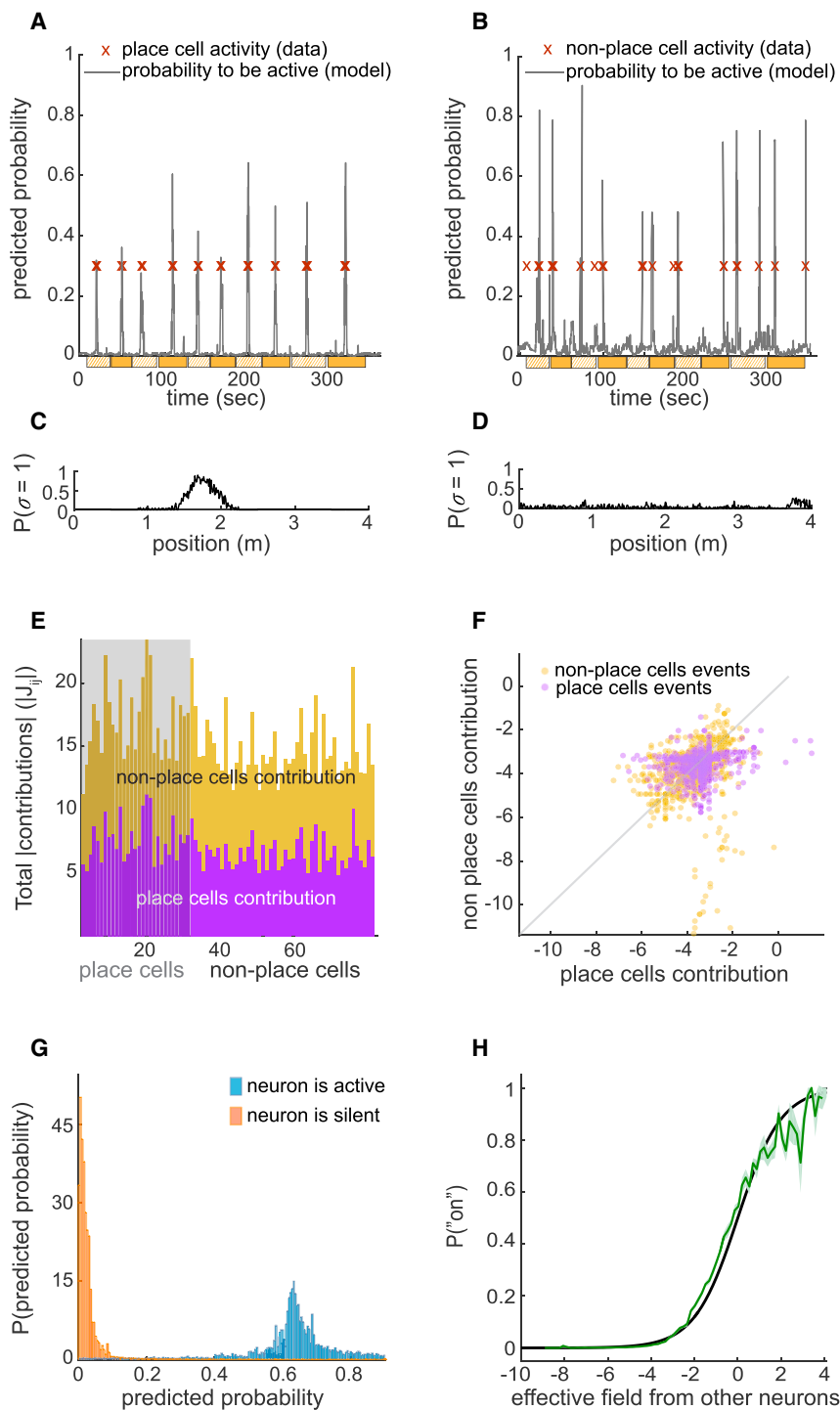


Figure 7. The Population Effective Field Predicts the Activity of Individual Neurons

(A) Predicted probability to be active versus time for a place cell. Red $\times$ marks show time points where the neuron was active. Each orange/stripped orange rectangle below the x axis indicates a trial.

(B) Predicted probability to be active versus time for a non-place cell. Red $\times$ marks show “1” bins in the data; i.e., time points where the neuron was active. Each orange/stripped orange rectangle below the x axis indicates a trial.

(C) Probability of activity versus position for the place cell depicted in (A).

(D) Probability of activity versus position for the non-place cell depicted in (B).

(E) Total magnitude of contributions, $|J_{ij}|$, from place and non-place cells to the activity of place/non-place cells based on the maximum entropy model. Each column represents the sum contributions from all cells to the activity of that individual neuron: the bottom purple part is the fraction that comes from place cells, and the top yellow part comes from non-place cells. The first shaded 32 columns are the place cells. Network contributions to both place and non-place cells seem to originate equally from the two sub-populations.

(F) Contributions from place cells and non-place cells to individual activity events of each neuron. Each point shown is a time point from a specific neuron (sub-sampled 1:1,000). The predicted probability includes a contribution from place cells (x axis) and from non-place cells (y axis). A point was colored purple when the time point belonged to a place cell and yellow if the time point belonged to a non-place cell. Network contributions to individual events along the epochs are balanced and originate from both groups of cells as well as the overall contributions (E).

(G) Predicted conditional probability distributions for all neurons to be active/silent for all time points. $P(\sigma_i = 1|h_{\text{eff}})$: time points where the neurons are active in the data colored in blue. $P(\sigma_i = 0|h_{\text{eff}})$: time points where the neurons are inactive, colored in orange.

(H) Probability of neuron to be active based on effective field. Shown is the relationship between the computed effective field and the probability of a neuron being active (green) compared with the parameter free prediction in Equation 12 (black). The shaded silhouette is the SD across.

To further assess how robust the model prediction is, we checked how dependent the prediction for each cell is on other individual neurons in the network by trying to omit one neuron at a time and quantifying the reduction in prediction quality. We found that no one cell omission has a significant effect on prediction quality. The scatterplot in Figure 8D shows that even the largest magnitude of contribution, $|J_{ij}|$, from a single cell is still

very small relative to the overall contribution from the rest of the cells, resulting in a very robust prediction that relies on the whole population and maintains its quality even when a few of the cells are ignored. Finally, the specific pattern of contributions from each cell has no clear correlation to the amount of spatial information of the cell or to its identity as a place/non-place cell, as shown in Figure S5.

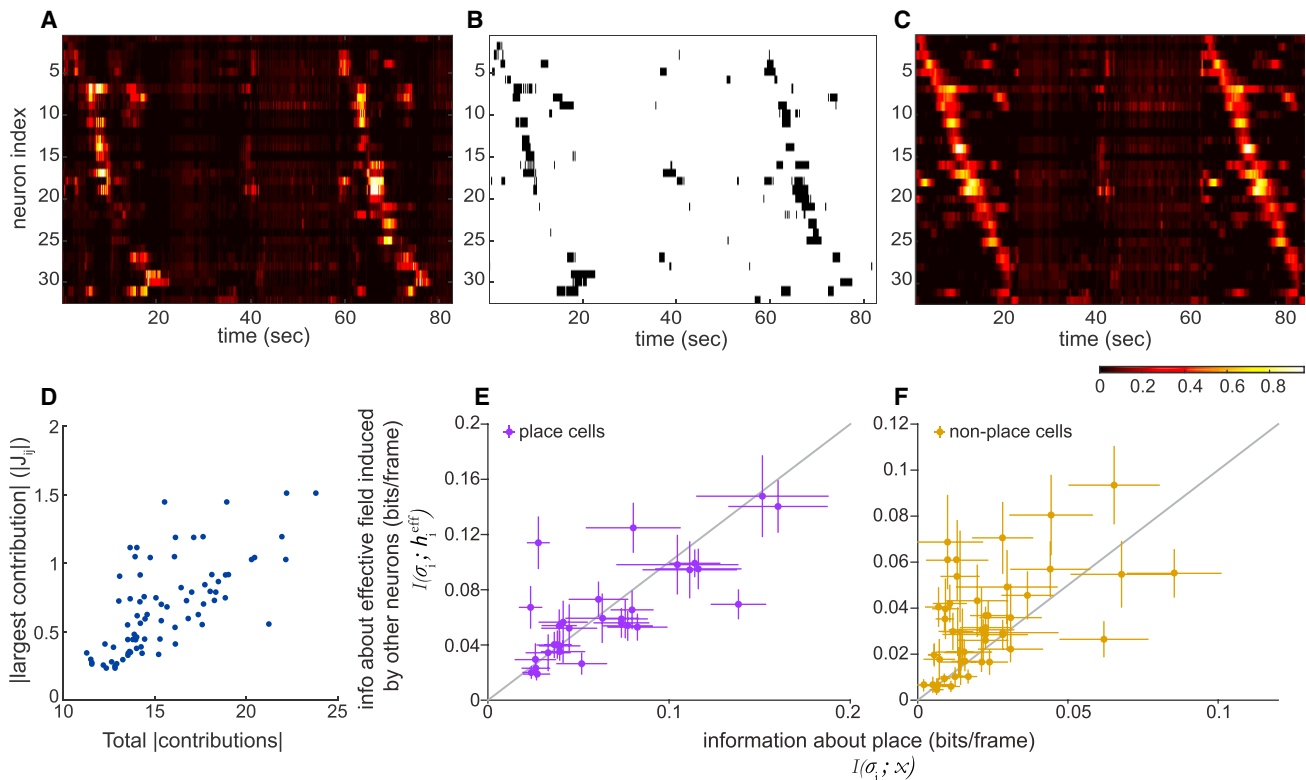


Figure 8. Collective and Place Information

(A) Predicted probability computed from the effective field in our model for all place cells along two consecutive runs along the linear track. The time window shown is the same as in (B) and (C). During the first run, cells 21–25 are predicted to “miss” their place field. Indeed, compared with the real data shown in (B), there is a missed field for these cells on that specific run, but all cells are predicted to be active in the second run.

(B) Real data of place cell activity during two runs down the linear track, corresponding to (A) and (C); note the missed events for cells 21–25 in the first run.

(C) Predicted probability computed from the independent place cells model for all place cells along two consecutive runs along the linear track as in (A). The predictions for the two runs are almost identical, with no indication of when fields should be missed.

(D) Largest magnitude of contribution, $|J_{ij}|$, from an individual cell versus the magnitude of total contributions from all cells. Each point represents a neuron. Even the largest contribution is small relative to the overall one; most cells rely on the whole network rather than on a small group of cells.

(E) Each point (with error bars for both axes) corresponds to the information the activity of a particular place cell carries about the state of the network versus the information carried about the position of the animal. Place cells carry similar amounts of information about the state of the network and about the animal’s position.

(F) As in (E) but for non-place cells. Non-place cells carry more information about the state of the network than about the position of the animal. Note that some of them carry non-negligible position information even though they do not have a sufficiently reliable localized place field to be classified as place cells.

DISCUSSION

High-Precision Agreement between Predictions from the Maximum Entropy Model and Experimental Observations in CA1

We have explored a description of activity in the hippocampus that focuses on the collective behavior of the network rather than on the sensory inputs or motor outputs. Extending a strategy that was originally applied to populations of neurons recorded in the retina, a “lower” sensory area (Ganmor et al., 2011a; Schneidman et al., 2006; Tkačik et al., 2010, 2014), we have built this model by taking from experiment a few basic quantities; in particular, the mean activity of each neuron and the correlations between activity in pairs of neurons. Beyond these constraints from experiments, we ask for a model that generates states of activity in the network that are as random as

possible so that the only structure is that required to match the constraints. This maximum entropy model is learnable from the data; given the size of the datasets to which we have access, there are no signs of over-fitting (Figure S1). The model is simple, but it passes a number of quite detailed tests.

Although the model is based on measurements of correlations between pairs of neurons, it makes predictions for the probability of any possible combination of activity and silence in the network. Although we cannot estimate the probabilities for each of these $2^{78} \sim 3 \times 10^{23}$ states, we can measure the probability that K out of the N neurons in the network are active simultaneously (Figure 5A), and we can measure the correlations among all 7.6×10^4 distinct triplets of neurons (Figure 6A), and in both cases, we see detailed, quantitative agreement between theory and experiment. The central quantity in our model is the “energy” or (negative) log probability, and we can predict the

distribution of this quantity across the network states that actually occur in the experiment, deep into the tail of rare states (Figure 5B). Finally, and perhaps most importantly, we can test the idea that activity in the network is collective by predicting the activity of one neuron based on the state of all the others (Figure 7). We have focused on the analysis of a single dataset, from 78 neurons in a 30-min recording session, but essentially the same results are found in 5 other datasets, as described in the [Method Details](#) of the [STAR Methods](#).

Our first conclusion then, is that the general strategy we have adopted succeeds in describing the distribution of activity across networks of ~ 80 pyramidal neurons. These results indicate that the maximum entropy method is a viable path for constructing global models of joint activity in central brain regions. We may yet find that the small quantitative discrepancies between theory and experiment are hints of things we are missing that become clearer in larger networks, as in the retina (Ganmor et al., 2011b; Tkačik et al., 2014), but the quantitative agreement we do find here sets a standard for what we should expect from such models. We emphasize that the correlation structures we see in this network are not simple: covariation among neurons cannot be captured by linear projection into a low dimensional space (Figure S6D), nor can the higher-order correlations be captured by a model of independent place cells, even allowing for arbitrarily complex and disconnected place fields (Figure 6C; Figure S3).

In our analysis, we investigated equal-time correlations; a reasonable next step is to incorporate temporal correlations into the model. Although recent studies have extended the maximum entropy approach to account for time dependencies (Mora et al., 2015), the dynamical aspect of the collective behavior of place and non-place cells remains a challenge.

Place and Non-place Cells Encode Information Collectively

Our results show that the activity of both place and non-place cells can be given a compact, unified description as part of a single network exhibiting collective behavior. All of the quantitative features of this collective behavior are predicted from the observed correlations between pairs of cells. We have seen that higher-order correlations are not predicted at all in a model where cells respond independently to the animal's location (Figure 6C; Figure S3C) and that there are even quantitative deviations of the pairwise correlations from this place field model (Figure S6C). Neither the pairwise correlations C_{ij} in the data nor the inferred interactions J_{ij} in our model are of low rank and the deviations of the correlations from the place field model are also not of low rank, suggesting that there is no global additive modulation of activity (Figure S6D). Place fields can shift from trial to trial (Fenton and Muller, 1998). In fact, discharge of place cells has long since been characterized as having variance beyond a simple location-dependent Poisson process. In addition, place cells exhibit other known background phenomena, such as the theta rhythm, which supports our observation that their firing is not independent and requires other potentially network-level effects to be considered to account for their full activity patterns (Barbieri et al., 2001; Fenton and Muller, 1998; O'Keefe and Recce, 1993; Olypher et al., 2002). However, if we

estimate these shifts from our data, then there is no obvious global pattern that can trivially explain the appearance of collective behavior (Figure S6F). We note that it is unlikely that our observations were caused by any fast timescale phenomena, such as theta-related modulation, because the relatively slow decay time of the calcium indicator dictates strong correlations between adjacent time bins. Consecutive identical 70-ms bins are effectively averaging over the entirety of theta.

In the view of network activity as a collective whole, we think of individual cells as following the state of the network, as summarized by the effective field (Equation 11). An alternative view is that each neuron is encoding information about the organism's behavior and its sensory inputs. In the hippocampus, the natural variable is the position of the mouse along the (virtual) track, and indeed, 32 of 78 of the cells in the population that we study here have clear place fields (Figure 2) but the majority of cells (46 of 78) do not. These non-place cells, whose activity is not significantly spatially modulated, could function as place cells in another environment, but they are far from being "silent" (Ferguson et al., 2011; Thompson and Best, 1989).

Previous efforts to accomplish reconstruction of an individual neuron's activity from the rest of the network made remarkable observations regarding the similarity between the population activity in anesthetized V1 and individual neurons' response to preferred orientation (Grinvald et al., 2003; Tsodyks et al., 1999). We note, however, that Grinvald et al. (2003) did not really demonstrate that the activity of single neurons was predictable, moment by moment, from the activity of other cells in the network, as we have done here.

One hypothesized mechanism that could explain why such a moment-by-moment prediction is possible is that pyramidal neurons in CA1 all receive spatially modulated inputs that span the whole environment, but the cells that exhibit clear fields are tuned to a specific range of the inputs, basically receiving them with a higher weight (C. Domnisoru and D.W.T., unpublished data). Such a structure of the system would also explain the fact that non-place cells seem to carry some spatial information and that the network encodes similar things over the full population. However, independent of its underlying mechanism, the fact that the network exhibits a nontrivial pattern of collective behavior that is captured by our relatively simple model does not seem to be an obvious consequence of known deviations from a simple place field model, and certainly such behaviors have not been predicted quantitatively by previous models. Because our model does not make reference to place, both classes of neurons appear on the same footing, but this leaves open the question of what really is being represented.

The Internal Network State Contributes Significantly to the Population Activity in CA1

The activity of individual place cells, by definition, carries information about position, and we can quantify this in bits: the information $I(\sigma_i; X)$ that the state of neuron i at one moment in time provides about the position of the mouse along the virtual track. Because the cells are active only in a few percent of the time frames, we expect this average, momentary information to be small in absolute terms, roughly ~ 0.1 bits, as seen in Figure 8E. But we can also ask how much information the activity of these

neurons provides about the effective field; that is, a summary of the state of the rest of the network, $I(\sigma_i; h_i^{\text{eff}})$. Figure 8E shows that, except for a few outliers, these two information measures are equal within error bars for each place cell.

Non-place cells are defined by the fact that there is no single place near which the probability of being active is high. Nonetheless, the activity in these cells can convey information about place, although typically less than the place cells, as we expect. For almost all of the non-place cells, however, we find that the information that activity provides about the effective field is much larger than the information about position, $I(\sigma_i; h_i^{\text{eff}}) \gg I(\sigma_i; x)$, as shown in Figure 8F. Beyond the conclusion that information shared collectively among neurons in the network is not entirely about place, these findings suggest that the activity should be understood as a combination of two significant contributors: the external variable, position, and the internal network state of the neurons. We suspect that it will be easier to discern the biological significance of this collectively encoded information when we can monitor larger populations of cells in a broader range of environments.

An interesting phenomenon observed experimentally in the dorsal hippocampus is “remapping,” which occurs when an animal is moved from one familiar environment to another, and the place cells in CA1 code for a different cognitive map in each environment. When discussing possible models for the hippocampal network, it is important to remember that these different maps partially overlap—a fact that has been difficult to resolve with the initial view of different spatial maps needing to be orthogonal to one another. This conflict has led to two possible mechanistic hypotheses: either we need a more sophisticated attractor map architecture than what has been proposed previously, or the hippocampus conjunctively encodes both map information and some other type of information (Skaggs and McNaughton, 1998). Our findings point toward the latter hypothesis.

Taken together, the evidence that the correlation patterns in the data only partially arise from place encoding and that additional information is encoded at the population level lead us to believe that, to understand CA1 neural activity, we should look beyond position encoding, into the interactions among neurons, which have equal or even greater influence on the observed activity patterns.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **CONTACT FOR REAGENT AND RESOURCE SHARING**
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 - Independent place cells model
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- Contributions from place and non-place neurons
- Examining the collective behavior of the network
- **DATA AND SOFTWARE AVAILABILITY**

SUPPLEMENTAL INFORMATION

Supplemental Information includes six figures and can be found with this article online at <https://doi.org/10.1016/j.neuron.2017.10.027>.

AUTHOR CONTRIBUTIONS

Modeling and analysis, L.M.; Data collection, J.L.G.; Writing – Original Draft, L.M. and W.B.; Writing – Review & Editing, L.M., J.L.G., C.D.B., D.W.T., and W.B.; Supervision, C.D.B., D.W.T., and W.B.

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REFERENCES

- Barbieri, R., Quirk, M.C., Frank, L.M., Wilson, M.A., and Brown, E.N. (2001). Construction and analysis of non-Poisson stimulus-response models of neural spiking activity. *J. Neurosci. Methods* **105**, 25–37.
- Battaglia, F.P., and Treves, A. (1998). Attractor neural networks storing multiple space representations: A model for hippocampal place fields. *Phys. Rev. E* **58**, 7738–7753.
- Bialek, W., Cavagna, A., Giardina, I., Mora, T., Silvestri, E., Viale, M., and Walczak, A.M. (2012). Statistical mechanics for natural flocks of birds. *Proc. Natl. Acad. Sci. USA* **109**, 4786–4791.
- Bialek, W., Cavagna, A., Giardina, I., Mora, T., Pohl, O., Silvestri, E., Viale, M., and Walczak, A.M. (2014). Social interactions dominate speed control in poising natural flocks near criticality. *Proc. Natl. Acad. Sci. USA* **111**, 7212–7217.
- Bullock, T.H. (1970). The reliability of neurons. *J. Gen. Physiol.* **55**, 565–584.
- Colgin, L.L., Leutgeb, S., Jezek, K., Leutgeb, J.K., Moser, E.I., McNaughton, B.L., and Moser, M.-B. (2010). Attractor-map versus autoassociation based attractor dynamics in the hippocampal network. *J. Neurophysiol.* **104**, 35–50.
- Dombeck, D.A., Harvey, C.D., Tian, L., Looger, L.L., and Tank, D.W. (2010). Functional imaging of hippocampal place cells at cellular resolution during virtual navigation. *Nat. Neurosci.* **13**, 1433–1440.
- Domnisoru, C., Kinkhabwala, A.A., and Tank, D.W. (2013). Membrane potential dynamics of grid cells. *Nature* **495**, 199–204.
- Droulez, J., and Berthoz, A. (1991). A neural network model of sensoritopic maps with predictive short-term memory properties. *Proc. Natl. Acad. Sci. USA* **88**, 9653–9657.
- Epsztein, J., Brecht, M., and Lee, A.K. (2011). Intracellular determinants of hippocampal CA1 place and silent cell activity in a novel environment. *Neuron* **70**, 109–120.
- Farvick, A., Dupont, L.M., and Eichenbaum, H. (2009). Distinct roles for dorsal CA3 and CA1 in memory for sequential nonspatial events. *Learn. Mem.* **17**, 12–17.
- Fenton, A.A., and Muller, R.U. (1998). Place cell discharge is extremely variable during individual passes of the rat through the firing field. *Proc. Natl. Acad. Sci. USA* **95**, 3182–3187.

- Ferguson, J.E., Jackson, J.C., and Redish, A.D. (2011). An inside look at hippocampal silent cells. *Neuron* 70, 3–5.
- Fuhs, M.C., and Touretzky, D.S. (2006). A spin glass model of path integration in rat medial entorhinal cortex. *J. Neurosci.* 26, 4266–4276.
- Ganmor, E., Segev, R., and Schneidman, E. (2011a). The architecture of functional interaction networks in the retina. *J. Neurosci.* 31, 3044–3054.
- Ganmor, E., Segev, R., and Schneidman, E. (2011b). Sparse low-order interaction network underlies a highly correlated and learnable neural population code. *Proc. Natl. Acad. Sci. USA* 108, 9679–9684.
- Gauthier, J.L., and Tank, D.W. (2017). Context-invariant encoding of reward location in a distinct hippocampal population. *bioRxiv*. <https://doi.org/10.1101/207043>.
- Gelbard-Sagiv, H., Mukamel, R., Harel, M., Malach, R., and Fried, I. (2008). Internally generated reactivation of single neurons in human hippocampus during free recall. *Science* 322, 96–101.
- Grinvald, A., Arieli, A., Tsodyks, M., and Kenet, T. (2003). Neuronal assemblies: single cortical neurons are obedient members of a huge orchestra. *Biopolymers* 68, 422–436.
- Hafting, T., Fyhn, M., Molden, S., Moser, M.-B., and Moser, E.I. (2005). Microstructure of a spatial map in the entorhinal cortex. *Nature* 436, 801–806.
- Hunsaker, M.R., Lee, B., and Kesner, R.P. (2008). Evaluating the temporal context of episodic memory: the role of CA3 and CA1. *Behav. Brain Res.* 188, 310–315.
- Jaynes, E.T. (1957). Information theory and statistical mechanics. *Phys. Rev.* 106, 620.
- Jaynes, E.T. (1982). On the rationale of maximum-entropy methods. *Proc. IEEE* 70, 939–952.
- Jensen, O., and Lisman, J.E. (1996). Hippocampal CA3 region predicts memory sequences: accounting for the phase precession of place cells. *Learn. Mem.* 3, 279–287.
- Káli, S., and Dayan, P. (2000). The involvement of recurrent connections in area CA3 in establishing the properties of place fields: a model. *J. Neurosci.* 20, 7463–7477.
- Kelemen, E., and Fenton, A.A. (2013). The Organization of Neuronal Discharge on Timescales of Milliseconds and Seconds Is Related to the Spatial Response Properties of Hippocampal Neurons. In *Advances in Cognitive Neurodynamics (III): Proceedings of the Third International Conference on Cognitive Neurodynamics 2011*, Y. Yamaguchi, ed. (Springer Netherlands: Dordrecht), pp. 421–427.
- Kesner, R.P., Hunsaker, M.R., and Gilbert, P.E. (2005). The role of CA1 in the acquisition of an object-trace-odor paired associate task. *Behav. Neurosci.* 119, 781–786.
- Kubie, J.L., and Muller, R.U. (1991). Multiple representations in the hippocampus. *Hippocampus* 1, 240–242.
- Langston, R.F., Ainge, J.A., Couey, J.J., Canto, C.B., Bjerknes, T.L., Witter, M.P., Moser, E.I., and Moser, M.-B. (2010). Development of the spatial representation system in the rat. *Science* 328, 1576–1580.
- Leutgeb, S., Leutgeb, J.K., Treves, A., Moser, M.-B., and Moser, E.I. (2004). Distinct ensemble codes in hippocampal areas CA3 and CA1. *Science* 305, 1295–1298.
- Levy, W.B. (1996). A sequence predicting CA3 is a flexible associator that learns and uses context to solve hippocampal-like tasks. *Hippocampus* 6, 579–590.
- Lisman, J.E. (1999). Relating hippocampal circuitry to function: recall of memory sequences by reciprocal dentate-CA3 interactions. *Neuron* 22, 233–242.
- Machens, C.K., Romo, R., and Brody, C.D. (2010). Functional, but not anatomical, separation of “what” and “when” in prefrontal cortex. *J. Neurosci.* 30, 350–360.
- Mante, V., Sussillo, D., Shenoy, K.V., and Newsome, W.T. (2013). Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature* 503, 78–84.
- Mazor, O., and Laurent, G. (2005). Transient dynamics versus fixed points in odor representations by locust antennal lobe projection neurons. *Neuron* 48, 661–673.
- McNaughton, B.L., Barnes, C.A., and O’Keefe, J. (1983). The contributions of position, direction, and velocity to single unit activity in the hippocampus of freely-moving rats. *Exp. Brain Res.* 52, 41–49.
- Mézard, M., Parisi, G., and Virasoro, M.-A. (1987). Replica symmetry breaking and the nature of the spin glass phase. In *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications*, M. Mezard, ed. (World Scientific Press), pp. 199–210.
- Moore, G.P., Perkel, D.H., and Segundo, J.P. (1966). Statistical analysis and functional interpretation of neuronal spike data. *Annu. Rev. Physiol.* 28, 493–522.
- Mora, T., Deny, S., and Marre, O. (2015). Dynamical criticality in the collective activity of a population of retinal neurons. *Phys. Rev. Lett.* 114, 078105.
- Mukamel, E.A., Nimmerjahn, A., and Schnitzer, M.J. (2009). Automated analysis of cellular signals from large-scale calcium imaging data. *Neuron* 63, 747–760.
- O’Keefe, J., and Conway, D.H. (1978). Hippocampal place units in the freely moving rat: why they fire where they fire. *Exp. Brain Res.* 31, 573–590.
- O’Keefe, J., and Dostrovsky, J. (1971). The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. *Brain Res.* 34, 171–175.
- O’Keefe, J., and Nadel, L. (1979). *Precis of O’Keefe & Nadel’s The hippocampus as a cognitive map*. *Behav. Brain Sci.* 2, 487–494.
- O’Keefe, J., and Recce, M.L. (1993). Phase relationship between hippocampal place units and the EEG theta rhythm. *Hippocampus* 3, 317–330.
- Olypher, A.V., Lánský, P., and Fenton, A.A. (2002). Properties of the extra-positional signal in hippocampal place cell discharge derived from the overdispersion in location-specific firing. *Neuroscience* 111, 553–566.
- Quiroga, R.Q., Reddy, L., Kreiman, G., Koch, C., and Fried, I. (2005). Invariant visual representation by single neurons in the human brain. *Nature* 435, 1102–1107.
- Redish, A.D., and Touretzky, D.S. (1999). Separating hippocampal maps. In *Spatial Functions of the Hippocampal Formation and the Parietal Cortex*, N. Burgess, K. Jeffery, and J. O’Keefe, eds. (Oxford University Press), pp. 203–219.
- Samsonovich, A., and McNaughton, B.L. (1997). Path integration and cognitive mapping in a continuous attractor neural network model. *J. Neurosci.* 17, 5900–5920.
- Sargolini, F., Fyhn, M., Hafting, T., McNaughton, B.L., Witter, M.P., Moser, M.-B., and Moser, E.I. (2006). Conjunctive representation of position, direction, and velocity in entorhinal cortex. *Science* 312, 758–762.
- Schmidt, M. (2007). UGM: A Matlab toolbox for probabilistic undirected graphical models. <https://www.cs.ubc.ca/~schmidtm/Software/UGM.html>.
- Schneidman, E., Berry, M.J., 2nd, Segev, R., and Bialek, W. (2006). Weak pairwise correlations imply strongly correlated network states in a neural population. *Nature* 440, 1007–1012.
- Shannon, C. (1948). A mathematical theory of communication. *Bell System Technical Journal* 27, 379–423.
- Skaggs, W.E., and McNaughton, B.L. (1998). Spatial firing properties of hippocampal CA1 populations in an environment containing two visually identical regions. *J. Neurosci.* 18, 8455–8466.
- Skaggs, W.E., McNaughton, B.L., Gothard, K.M., and Markus, E.J. (1992). An information-theoretic approach to deciphering the hippocampal code. In *Proceedings of the 5th International Conference on Neural Information Processing Systems* (Morgan Kaufmann Publishers Inc.), pp. 1030–1037.
- Thompson, L.T., and Best, P.J. (1989). Place cells and silent cells in the hippocampus of freely-behaving rats. *J. Neurosci.* 9, 2382–2390.
- Tkačik, G., Schneidman, E., Berry, I., Michael, J., and Bialek, W. (2006). Ising models for networks of real neurons. *arXiv*. [arXiv:q-bio/0611072](https://arxiv.org/abs/q-bio/0611072).

- Tkačik, G., Schneidman, E., Berry, I., Michael, J., and Bialek, W. (2009). Spin glass models for a network of real neurons. *arXiv*. arXiv:0912.5409.
- Tkačik, G., Prentice, J.S., Balasubramanian, V., and Schneidman, E. (2010). Optimal population coding by noisy spiking neurons. *Proc. Natl. Acad. Sci. USA* 107, 14419–14424.
- Tkačik, G., Marre, O., Mora, T., Amodei, D., Berry, M.J., II, and Bialek, W. (2013). The simplest maximum entropy model for collective behavior in a neural network. *J. Stat. Mech.* 2013, P03011.
- Tkačik, G., Marre, O., Amodei, D., Schneidman, E., Bialek, W., and Berry, M.J., 2nd (2014). Searching for collective behavior in a large network of sensory neurons. *PLoS Comput. Biol.* 10, e1003408.
- Treves, A., and Rolls, E.T. (1992). Computational constraints suggest the need for two distinct input systems to the hippocampal CA3 network. *Hippocampus* 2, 189–199.
- Tsodyks, M., and Sejnowski, T. (1995). Associative memory and hippocampal place cells. *Int. J. Neural Syst.* 6, 81–86.
- Tsodyks, M., Kenet, T., Grinvald, A., and Arieli, A. (1999). Linking spontaneous activity of single cortical neurons and the underlying functional architecture. *Science* 286, 1943–1946.
- Waydo, S., Kraskov, A., Quiñ Quiroga, R., Fried, I., and Koch, C. (2006). Sparse representation in the human medial temporal lobe. *J. Neurosci.* 26, 10232–10234.
- Wills, T.J., Cacucci, F., Burgess, N., and O’Keefe, J. (2010). Development of the hippocampal cognitive map in preweanling rats. *Science* 328, 1573–1576.
- Yartsev, M.M., Witter, M.P., and Ulanovsky, N. (2011). Grid cells without theta oscillations in the entorhinal cortex of bats. *Nature* 479, 103–107.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental Model Organisms		
Mouse Thy1-GCaMP3-WPRE	Janelia GENIE	GP2.11
Software and Algorithm		
Data acquisition with ScanImage	Vidrio Technologies	https://vidriotechnologies.com/
Custom-written MATLAB code for modeling and analysis	MATLAB	https://www.mathworks.com/

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Leenoy Meshulam (leenoy@princeton.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

Data included unpublished neural recordings from transgenic male mice C57BL/6J-Tg (Thy1-GCaMP3) GP2.11Dkim/J (Jackson Labs, stock number: 028277). At the time of recording, mice ranged between 3 and 5 months in age. Mice and their littermates were housed together until surgical implantation of the optical window. After implantation, mice were individually housed in transparent cages on a reverse light cycle, with behavioral sessions occurring during the dark phase. All techniques were approved by the Institutional Animal Care and Use Committee at Princeton University.

METHOD DETAILS

Experimental Methods

Surgery

Optical access to the hippocampus was obtained as previously described (Dombeck et al., 2010). Briefly, a ~3mm circular hole was cut into the skull centered at 1.8 mm lateral and 2.0 mm posterior to bregma on the left side of the skull. The cortical tissue overlying the hippocampus was aspirated, and a circular metal cannula with a #1 coverslip affixed to the bottom was implanted, with a thin layer of Kwik-sil (WPI) between the hippocampus and coverslip.

Virtual reality setup

Water-restricted mice were trained to receive water rewards by running on a 4 m virtual linear track using a virtual reality setup as described in (Gauthier and Tank, 2017). Mice ran on an air-supported Styrofoam treadmill constrained to rotate only around the horizontal axis (pitch) to prevent turning. Treadmill movement was read out using an optical mouse, and a visual display of the virtual environment was projected onto a 270° toroidal screen. Images of neural activity were acquired using two photon laser scanning microscopy as previously described (Dombeck et al., 2010).

Data Processing

Image processing

Images were analyzed using toolboxes and custom scripts written in MATLAB. Acquired frames were corrected for brain motion by aligning to a template image, and cell finding was performed on the corrected movie using a modified version of the PCA-ICA algorithm (Mukamel et al., 2009), followed by manual human annotation for verification.

From continuous fluorescence signal to a binary time series

To convert the neural activity reported by the kinetics of the GCaMP3 indicator, we used independent exponential fits for the rise and decay time of the fluorescence signal for each activity event. After identifying all maxima that can potentially be activity events, as described in the main text, the rising phase of each response was fit with a single exponential function. Traces whose time course deviated more than 3σ from average were discarded, since they usually reflected residual motion artifacts or external noise sources. Next, the decay trace was fit to a double exponential $f(t) = A_0 + A_1 e^{-(k_1 t)} + A_2 e^{-(k_2 t)}$. In the case that k_2 was slower than 1σ from the average trace for that cell, the time point was assigned a value of 1, as it was assumed the cell kept on firing and the decay slope was

fit again from the next time point and on, repeating until the slope was within 1σ from the average. In the case the trace decayed faster than 1σ from the average for that neuron, the time point was assigned a 0 value.

To explore the sensitivity of our results to the binarization procedure and the decay time of the calcium indicator, we re-evaluated our model results for gradually increasing values of the threshold, starting at the original 3.5σ value (indicated in Figure 1D). The overall structure of correlations stayed remarkably similar as the threshold increased, though they weakened. At a threshold of $\sim 3 * 3.5\sigma$, correlation values dropped to half their original value, with a similar overall structure. However, even for this higher threshold all model predictions and results still held. We started observing a decrease in the model's predictive power only when the threshold was set to a value so high that a non-negligible number of events were missed completely. Thus, even if all correlation values measured through calcium signal are scaled up somewhat because of the tails of decay time, it does not have a meaningful impact on our results.

Maximum Entropy Model

Learning the model

To build the maximum entropy models constrained by the firing rates and covariances computed from the data, we performed Markov Chain Monte Carlo (MCMC). In every cycle, we shift the values of our model parameters $\{h_i, J_{ij}\}$ in the direction of the gradient toward the solution. The log-linear form of the Boltzmann distribution guarantees us convergence for any set of observed parameters $\{h_i, J_{ij}\}$. The algorithm we use leverages quasi-Newton approximations, which estimate the Hessian of the target distribution from previous samples and gradients generated by the sampler (Schmidt, 2007). To ensure our final solution is not sensitive to initial conditions, we incorporated a long enough burn-in period (usually 100,000 cycles) where we threw away the samples before starting to record them. The estimate for our model parameters $\{h_i, J_{ij}\}$, which serve as the coefficients of the observables in our cost function, lets us obtain the full probability distribution for the joint activity of the neurons. Once we have the distribution, we can sample synthetic data from it, i.e., simulate a set of configurations (population states), according to their probability to occur in our model. Our probability distribution assigns a probability value for any possible configuration. To test for overfitting, we used 3/4 of trials (randomly selected) to train the model, and left out a 1/4 as a test set. Figure S1 shows the correlation values predicted by the model as computed from the sampled population states, against the ones computed from the test set. We can see they are in agreement within error bars.

Survey of additional experiments

We analyzed 6 datasets recorded in CA1 of 4 GCaMP3 transgenic mice. The number of neurons in the datasets ranged 21-78: 1. $N=78$ (as described in the main text), 2. $N=69$, 3. $N=55$, 4. $N=68$, 5. $N=46$, 6. $N=40$. Each session lasted $T \sim 25$ minutes, where each run along the track lasted $K \sim 15$ sec. A separate model was constructed for each session. Performance of the model for these datasets is summarized in Figure S2. Only 4 of the datasets shown for ease of visualization.

Independent place cells model

We have explored a model in which the states of neural activity are a coherent, collective whole, with each cell's probability of being active determined by the activity of the other cells in the network. This model makes no explicit reference to external inputs, although these may be encoded implicitly through the pattern of correlations. A very different approach is to imagine that each neuron is "responding" independently to the relevant sensory stimuli, and since we are in the hippocampus the natural stimulus variable is the (virtual) location of the mouse. Indeed, we made use of intuition from this model when we rationalized the pattern of pairwise correlations, arguing that the negative correlations arise because cells representing different places in the environment cannot be active simultaneously.

The idea of tracing correlations in neural activity back to correlations in their inputs is very old, dating back to a time when it was hoped that analysis of correlations could provide an unambiguous measure of anatomical connectivity (Bullock, 1970; Moore et al., 1966). The decomposition of correlations into stimulus-driven ("signal") and intrinsic ("noise") components remains popular in the analysis of more peripheral sensory systems. The standard approach in those cases is to present precise repetitions of the sensory input, and compare the correlations among neurons computed from properly aligned trials with those computed after the trials have been shuffled, leaving only the stimulus – driven correlations. Importantly, this is an experiment we can perform as observers of the brain, but not a distinction that is available to the brain itself. Nonetheless it is interesting to ask how model of neurons independently encoding place would compare to the maximum entropy model of collective activity that we have considered here.

Following conventional methods, we can estimate the probability that each neuron will be in the "on" state during the time bin when the mouse visits a particular place along the virtual track as in Figure 2C in the main text. There we see that for 32/78 neurons, there is a clear position near which the probability of being active is very high, even close to one, with activity having near zero probability away from this point; these are classical place fields. For the remaining 46/78 cells, however, the peak probability of being active is not very high, the activity is much more diffused, and in several cases, there are multiple low peaks. We have referred to these as "non-place cells" in the main text, although it is clear that they carry some spatial information, even if not conforming to the classical picture of place cells.

Let us refer to the place field of cell i as $F_i(x)$; this is the probability that cell i will be active in a time bin when the mouse is at virtual position x . If all cells are responding to the position variable independently, then we can generate synthetic sequences of activity just by reference to these probabilities as they vary along the trajectory $x(t)$. Further, we can predict the expectation values of the binary variables $\{\sigma_i\}$ and their correlations:

$$\langle \sigma_i \rangle_{\text{pl}} = \int dx P(x) F_i(x), \quad (13)$$

$$\langle \sigma_i \sigma_j \rangle_{\text{pl}} = \int dx P(x) F_i(x) F_j(x), \quad (14)$$

$$\langle \sigma_i \sigma_j \sigma_k \rangle_{\text{pl}} = \int dx P(x) F_i(x) F_j(x) F_k(x), \quad (15)$$

where the subscript pl reminds us that these predictions arise from a place field model and $P(x)$ is the probability of finding the mouse at virtual position x .

In [Figure S3A](#) we show the predictions for the pairwise correlations,

$$C_{ij}^{\text{pl}} = \langle \sigma_i \sigma_j \rangle_{\text{pl}} - \langle \sigma_i \rangle_{\text{pl}} \langle \sigma_j \rangle_{\text{pl}}, \quad (16)$$

compared with what we see experimentally.

The predictions from an independent place field model do a reasonable job of reproducing the observed pairwise correlations, although we will see that the spectrum of the full matrix is not well described ([Figure S6D](#)). The picture changes substantially when we try to predict the correlations among triplets of neurons, as we did with the maximum entropy model in [Figure 6](#) in the main text. In [Figure S3B](#) we show the triplet correlations predicted by the model of independent place fields, compared with the experimental data. This is a more detailed version of the green line in [Figure 6C](#) in the main text.

The model of independent place fields significantly underestimates the correlations among triplets of neurons, so that a plot of predicted versus observed C_{ijk} has a slope of $\sim 1/2$ across the full dynamic range ([Figure S3B](#)). But if we zoom in on the most common, small correlations, we see that the situation is much worse. Throughout the window shown in [Figure S3C](#), the model of independent place field predicts essentially zero triplet correlations.

Defining place cells

In order to identify which neurons qualify as place cells, we first divided the virtual 4-m-long track into 20 spatial bins and summed the number of “on” moments every neuron had in each bin. Following the strategy described in ([Skaggs et al., 1992](#)), to categorize how good a description of a location is given by an individual neuron, we measured the extent to which the information is concentrated in a specific place. We assigned 5 spatial bins as the range for a potential place field. The neuron’s mean activity (over all trials) center of mass determined the location field’s center bin. Spatial bins of the place field were defined as the center bin and 2 more bins from each side. Next, we implemented a two-step approach as described in previous studies. First, we computed the spatial information for each neuron per place field ([Domnisoru et al., 2013](#); [Skaggs et al., 1992](#); [Yartsev et al., 2011](#))

$$I = \sum_{k=1}^5 p_k \frac{\langle \sigma_i \rangle_k}{\langle \sigma_i \rangle} \log_2 \frac{\langle \sigma_i \rangle_k}{\langle \sigma_i \rangle} \quad (17)$$

where p_k is the probability of the mouse to be in the k -th bin of the place field, $\langle \sigma_i \rangle_k$ is the mean activity of neuron i in the k -th bin of the place field, $\langle \sigma_i \rangle$ is the overall mean activity and the index k runs over the spatial bins of the place field. Note that if we were to extend this calculation to all bins, we would get the measure for any place information in the neural signal (not necessarily localized), $I(\sigma_i; x)$, described in the main text. The second step in our two-step approach is to perform a shuffling test ([Langston et al., 2010](#); [Wills et al., 2010](#)), shifting the entire sequence of activity for every neuron in each trial by a random interval. The end of the run was wrapped to the start to create a circular permutation, and the interval was picked at random from a uniform distribution of all durations between 3 s to the trial’s duration. We repeated the process 500 times per neuron and computed the spatial information each time. This procedure has the advantage of effectively decoupling the neural activity from the mouse’s location, while not breaking its temporal structure. A neuron qualified as a place cell if its information per active time bin (“on” moment) was at least in the 80th percentile of the shuffled distribution for that neuron. In the case of a neuron with more than one field, the procedure was performed for each field separately and success in either of them would have qualified the neuron to be a place cell (rare). Out of the 78 neurons in the dataset reported in the main text, 32 qualified as place cells according to the shuffling criterion. This fraction was similar in the other 5 datasets we analyzed, ranging from 28% to 41%.

Contributions from place and non-place neurons

To determine whether there exists a specific structure of contributions to the predicted probability to fire ([Figure 7](#)) which differs between place and non-place cells, we examined the positive and negative contributions that each neuron receives. It seems that the structure of network contributions varies with no relationship to the cell’s identity or to the amount of spatial information the cell holds. Supplementary figure S5 shows the contributions, $|J_{ij}|$, to 6 example cells: 3 place cells (left column) and 3 non-place cells (right column). The three cells shown in each column were chosen such that they have similar value of spatial information. Despite that, different cells show different patterns of contributions from the rest of the place and non-place cells in the population.

The 1st place cell on the left has a contributions pattern we would have expected from a place cell – large positive contributions from the two place cells which have fields before and after it and negative contributions from the rest of cells. However, the other two neurons in the same column, though possess similar amounts of spatial information, show a more complicated pattern of network contributions. The second neuron has positive contribution from place cells that code for very different locations along the environment than its own, as well as from many of the non-place cells. The last place cell has an expected large contribution from the place cell right next to it, but also multiple positive contributions from non-place cells. The same unexpected variance of contribution patterns is characteristic for the three non-place cells on the right. While one has only negative contributions from all cells, the other two show both positive and negative contributions spread across place and non-places groups. In conclusion, the pattern of contributions from the cells in the network is indicative of the cell's class, nor of how much spatial information it carries.

Examining the collective behavior of the network

To further explore potential explanations that could account for the covariation of the neurons that we observe, since place-coding alone does not, we looked into two known sources of variability in CA1 neural activity: global changes in the network excitation and trial-by-trial changes in place field locations.

First, we ask whether the global pattern of correlations among pairs of neuron (Figure S6A) might be the result of activity in the network being confined to a space with dimensionality much lower than the number of neurons, as has been suggested in other systems (Machens et al., 2010; Mante et al., 2013; Mazor and Laurent, 2005). To test this possibility, we compute the eigenvalues of the covariance matrix C_{ij} as defined in Equation 2 in the main text, with the results shown in blue in Figure S6D. The covariation among neurons cannot be captured by linear projection into a low dimensional space, nor can the higher-order correlations be captured by a model of independent place cells, even allowing for arbitrarily complex and disconnected place fields (Figure S3C and Figure 6C in the main text). Moreover, the eigenvalue spectrum of the difference in correlations between the data and the place-related activity shows no sign of being low rank either (Figure S6D, in red), indicating that no simple model for global modulation of the network that we might attempt to write down could successfully capture it.

Next, we turn to the trial-by-trial changes in place field locations. This instability of the place cells' code, originally referred to as "excess firing variance" (Fenton and Muller, 1998), is characterized by changes in discharge during different passes of the animal through the same location. In the positional domain, place fields occasionally get dropped or even missed completely (e.g., Figure 8B in the main text). We explored the changes in their locations by comparing trials. In Figure S6E we show all place fields location changes, δL_{ii} :

$$x_i^{CM} = \frac{\sum_t X(t) \sigma_i(t)}{\sum_t \sigma_i(t)} \quad (18)$$

$$\delta L_{ii} = x_{ii}^{CM} - \langle x_i^{CM} \rangle. \quad (19)$$

For neuron i in any trial l , x_{ii}^{CM} is the place cell's activity's center of mass in that trial, and $\langle x_i^{CM} \rangle$ is the center of mass averaged across all trials ("true location"). Decomposition of the location shifts' matrix (Figure S6E) reveals no single pattern or global mode that could account for the collective behavior of the system.

We conclude that none of the global phenomena we tested accounts for the collective nature of the system. Indeed, we have not ruled out the possibility that there is a different variable of a more complex nature that is nonlinearly embedded in the data and could capture the observed covariation beyond position coding. However, since neither the covariance matrix of the data itself, nor the difference between this matrix to the covariance matrix of place-related activity only, is low rank, no single linear variable could help explain the data simply by addition to the low dimensional feature of place coding.

DATA AND SOFTWARE AVAILABILITY

Custom MATLAB modeling and analysis code as well as data will be made available upon request (fulfilled by Lead Contact, leenoy@princeton.edu).